

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q13-0237

Tuckasegee Tie 138 kV  
May 24, 2015

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0237. The Interconnection Customer proposes a 866.0 MW natural gas combined cycle facility interconnecting at the Tuckasegee Tie 138 kV substation in the Virginia area. The request was received on May 16, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 135 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$101,919,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

The TUCKASEGEE TIE – ELECTRIC AVENUE FACILITY 138.0 kV is overloaded to 295.9% of its 205 MVA emergency rating for the loss of the MERCED – TUCKASEGEE TIE 138.0 kV. The project contributes 66.2% DFAX (573.7 MW) to this overload.

MERCED – TUCKASEGEE TIE 138.0 kV line: 221.6% loading (235 MVA) following the outage of the TUCKASEGEE TIE – ELECTRIC AVENUE FACILITY 138.0 kV Line; DFAX = 42.5%.

MERCED – SILVER SPUR SUBSTATION 138.0 kV Line: 218.2% loading (228 MVA) following the outage of the ELECTRIC AVENUE FACILITY – TUCKASEGEE TIE 138.0 kV; DFAX = 42.5%.

ABILENE MAPLE ST – RED RIVER 138.0 kV L: 216.8% loading (205 MVA) following the outage of the ELECTRIC AVENUE FACILITY – TUCKASEGEE TIE 138.0 kV; DFAX = 42.5%.

The SILVER SPUR – ABILENE MAPLE ST 138.0 kV L is overloaded to 215.4% of its 222 MVA emergency rating for the loss of the ELECTRIC AVENUE FACILITY – TUCKASEGEE TIE SUBSTATION 138.0 kV Line. The project contributes 42.5% DFAX (368.5 MW) to this overload.

ELECTRIC AVENUE FACILITY – RIVERTON 230 138.0 kV: 198.1% loading (224 MVA) following the outage of the TUCKASEGEE TIE – MERCED 138.0 kV; DFAX = 41.4%.

RED RIV – LIBBY 138.0 kV: 175.8% loading (219 MVA) following the outage of the ELECTRIC AVENUE FACILITY – TUCKASEGEE TIE 138.0 kV; DFAX = 33.3%.

For the loss of MERCED – TUCKASEGEE TIE 138.0 kV, the RIVERTON 230 – REMINGTON CT 138.0 kV loads to 136.5% of its 240 MVA rating. Project DFAX: 25.0%.

TUCKASEGEE TIE – HOGAN RD 138.0 kV: 129.3% loading (247 MVA) following the outage of the TUCKASEGEE TIE SUBSTATION – ELECTRIC AVENUE FACILITY 138.0 kV; DFAX = 57.5%.

The LIBBY – CONTRA COSTA 138.0 kV is overloaded to 129.0% of its 296 MVA emergency rating for the loss of the ELECTRIC AVENUE FACILITY – TUCKASEGEE TIE 138.0 kV. The project contributes 25.8% DFAX (223.4 MW) to this overload.

The MILLER – CONTRA COSTA 138.0 kV Cir. 1 is overloaded to 108.6% of its 329 MVA emergency rating for the loss of the MILLER – CONTRA COSTA 138.0 kV Ckt 2. The project contributes 10.1% DFAX (87.3 MW) to this

overload.

The LOUISA GENERATION FACILITY SUBSTATION 345/138kV TX is overloaded to 106.8% of its 462 MVA emergency rating for the loss of the FLUVANNA P.S – LOUISA GENERATION FACILITY 345.0 kV line. The project contributes 14.2% DFAX (122.6 MW) to this overload.

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                           | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------------|---------------------------------------------|---------------|-----------------------|
| TUCKASEGE TIE – ELECTRIC AVENUE FACILITY 138.0 kV  | Rebuild line with higher-capacity conductor | \$3,391,000   | 100% (\$3,391,000)    |
| TUCKASEGEE TIE – MERCED 138.0 kV                   | Rebuild line with higher-capacity conductor | \$13,653,000  | 100% (\$13,653,000)   |
| MERCED – SILVER SPUR SUBSTATION 138.0 kV Line      | Rebuild line with higher-capacity conductor | \$14,109,000  | 100% (\$14,109,000)   |
| ABILENE MAPLE STREET – RED RIV 138.0 kV            | Rebuild line with higher-capacity conductor | \$3,363,000   | 100% (\$3,363,000)    |
| SILVER SPUR – ABILENE MAPLE ST 138.0 kV L          | Rebuild line with higher-capacity conductor | \$4,236,000   | 100% (\$4,236,000)    |
| ELECTRIC AVENUE FACILITY – RIVERTON 230 138.0 kV   | Rebuild line with higher-capacity conductor | \$6,044,000   | 100% (\$6,044,000)    |
| RED RIV – LIBBY 138.0 kV                           | Rebuild line with higher-capacity conductor | \$2,334,000   | 100% (\$2,334,000)    |
| RIVERTON 230 – REMINGTON CT 138.0 kV               | Rebuild line with higher-capacity conductor | \$27,080,000  | 75% (\$20,306,000)    |
| HOGAN RD – TUCKASEGEE TIE SUBSTATION 138.0 kV      | Rebuild line with higher-capacity conductor | \$16,615,000  | 100% (\$16,615,000)   |
| LIBBY – CONTRA COSTA 138.0 kV                      | Rebuild line with higher-capacity conductor | \$6,452,000   | 100% (\$6,452,000)    |
| MILER – CONTRA COSTA 138.0 kV                      | Reconductor line                            | \$5,956,000   | 100% (\$5,956,000)    |
| LOUISA GENERATION FACILITY SUBSTATION 345/138kV TX | Replace transformer with higher-rated unit  | \$13,273,000  | 41% (\$5,460,000)     |
|                                                    | Total allocated to project                  |               | \$101,919,000         |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.